

Last-Mile Bottleneck



Token-Based Printing

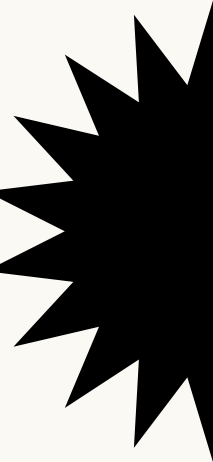
Enhancing Efficiency in Campus Workflow

Student Workflow

The student workflow utilizes a streamlined token-based system, allowing students to submit print jobs asynchronously. Upon submission, a unique 5-digit token is generated, providing students with real-time tracking of their orders. This process reduces wait times and enhances user satisfaction, making printing more efficient and accessible to all campus users.

Operator Workflow

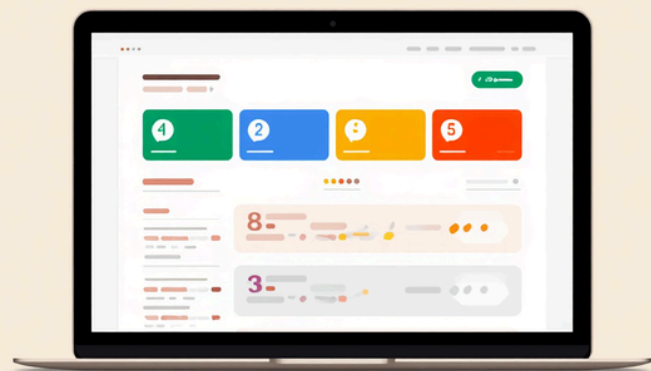
The operator workflow is designed to optimize printing operations by managing print jobs through a centralized dashboard. Operators can easily monitor incoming tokens and prioritize tasks based on urgency and resource availability. This method minimizes chaos and ensures that all print jobs are completed promptly, improving overall operational efficiency on campus.



Platform Interface Demo

Student Dashboard

User-friendly interface for seamless uploads



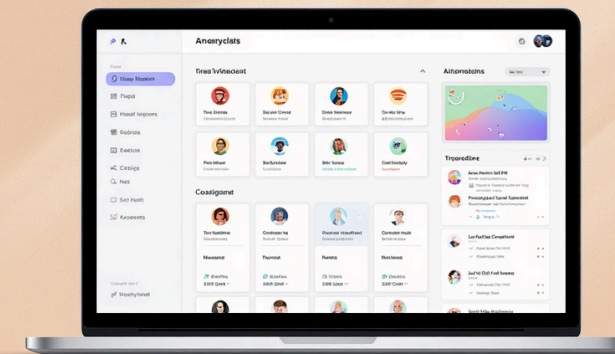
Token Generation

Efficient token creation for print jobs



Admin Dashboard

Comprehensive management and print monitoring



PDF Upload UI

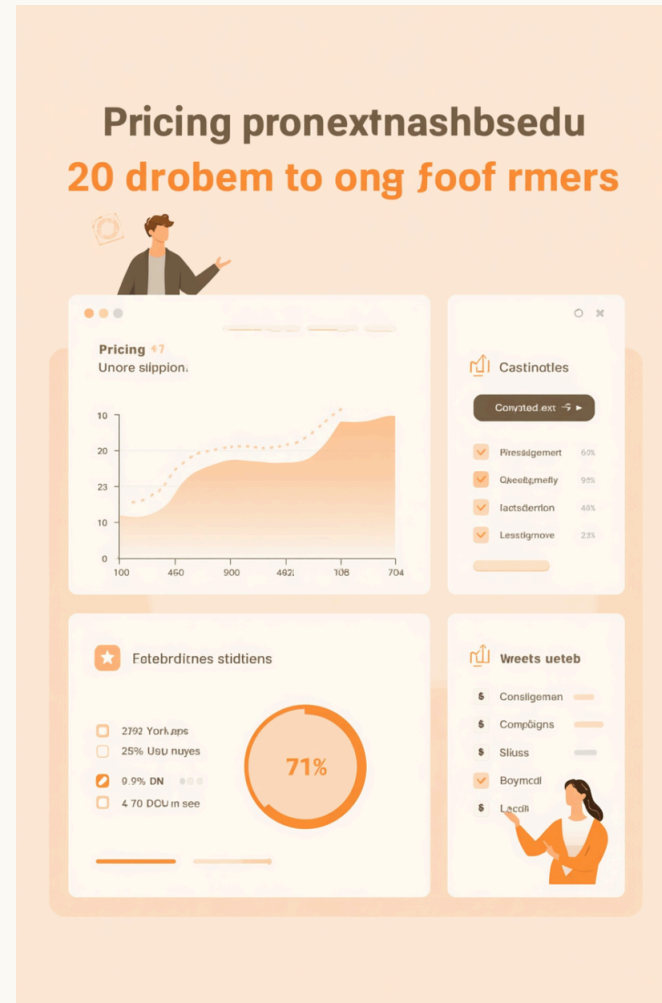
Quick and straightforward document submissions





Automated PDF

Efficiently divides documents into manageable parts



Dynamic Pricing

Adapts costs based on real-time factors



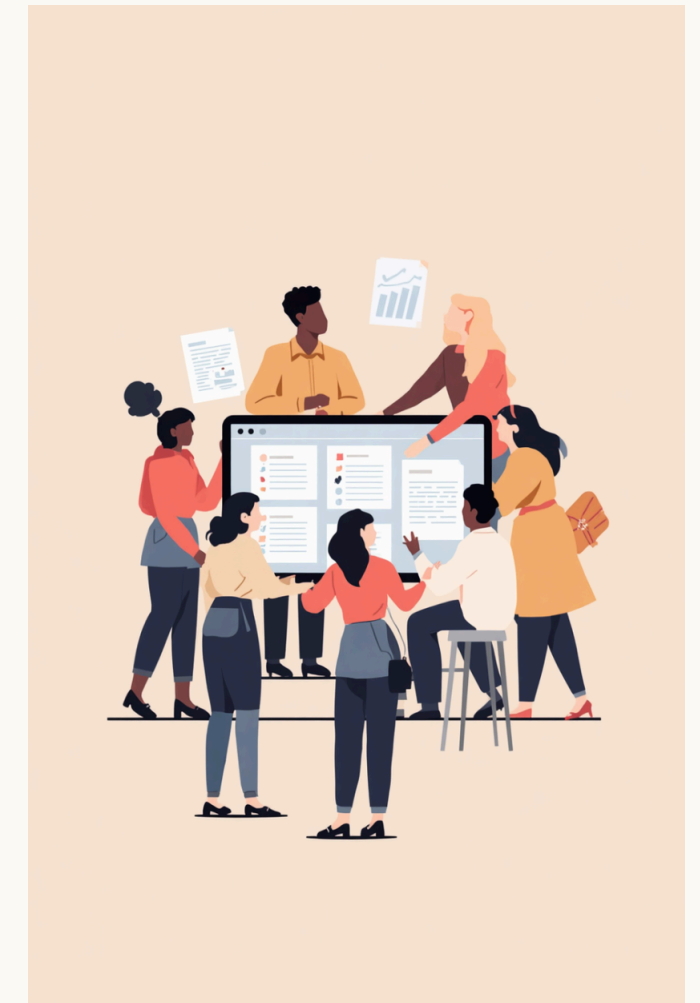
Secure Token

Ensures safe document access and printing



Cohesive Workflow

Integrates solutions for optimal efficiency



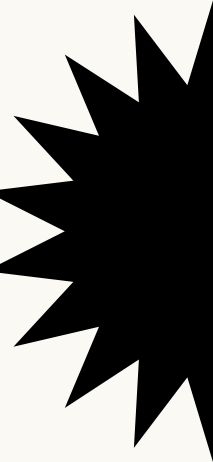
Collaborative Effort

Enhances teamwork in document management

Core Features and Workflow Overview

Architecture & Technology

Overview of System Components



Frontend Technologies

The frontend utilizes **React.js** for a dynamic user experience, enhanced by **Firebase Authentication** for secure logins and **Vite** for rapid development and build processes.

Backend Framework

The backend is powered by **FastAPI** (Python), enabling efficient API requests, alongside **PyPDF** for document processing and **MongoDB Atlas** for scalable data storage solutions.

Cloud Infrastructure

Utilizing **Firebase Cloud Storage**, the system ensures reliable document storage and seamless access, supporting asynchronous operations and enhancing user workflow for both students and administrators.